

Assignment 4

Due Date: December 10, 2010 by 5:30 pm

1. (20 points) This assignment will deal with modeling a simple harmonic oscillator such as that of a mass on a spring without any driving force on a frictionless surface. In this case, the net force on the system can be modeled using Newton's second law of motion. This yields the following relationship

$$F = ma = \frac{d^2 x}{dt^2} = -(m\omega^2)x$$

The equations of motion can be simplified further to two linear ordinary differential equations

$$\frac{dv}{dt} = -\frac{kx}{m}$$

$$\frac{dx}{dt} = v$$

- a. Create a function in octave that will return a matrix with two columns whose values are these two derivatives. Use the ODE functionality in octave to integrate the equations of motion and get the positions and velocities at different times t . To do so you will need a function called `ode23` (or `ode45`, which is a little more accurate), which is an add-on to octave. Get a copy online and put it in the directory where you are doing your octave work. To use it, call it as if it was a built-in function.
- b. Plot the position of the mass as a function of time. You will need to supply initial conditions and a simulation interval: choose $x_0=0$, $v_0 = 1$ and an interval of 20, with a mass of 534 g and a spring constant of 57 N/m.